Pedro Fonseca

Full Time Lecturer Computer Science Department - UTRGV
PhD Candidate in Computer Science — UTRGV
Machine Learning — Lean Construction — Decision Support Systems

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Profile

Full-time faculty member teaching courses including Software Engineering, Game Development, and Computer Science Foundations. Develop and deliver hands-on coursework covering programming, system design, and applied computing concepts.

PhD Candidate in Computer Science specializing in Machine Learning applications for construction and infrastructure systems. Research focuses on contractor performance prediction using data-driven methods, Lean Construction, and decision-support systems.

Education

PhD in Computer Science (Interdisciplinary Applications) 2024–Present
University of Texas Rio Grande Valley
Emphasis: Machine Learning, Lean Construction

M.Sc. in Intelligent Systems 2016–2017
Tecnológico de Monterrey
Emphasis: Data Science

B.A. in Digital Arts 2009–2013
Tecnológico de Monterrey
Emphasis: Human–Computer Interaction

Research & Projects

Contractor Performance Prediction using ML and WBS Similarity
Developed machine learning models integrating contractor historical data with work-package semantic similarity (Sentence Transformers). Achieved strong predictive performance ($R^2 \sim 0.5$) using XGBoost models.

Lean Construction Quality Audit System (Dos Bocas Refinery)
Designed and deployed a digital audit platform supporting 5,000+ inspections, 80+ contractors, and multi-year project lifecycle.

UAV-based Biomass Estimation using Photogrammetry
Developed CHM-based segmentation pipeline using watershed algorithms for biomass and carbon estimation.

Publications

- Fonseca-Ortiz, P., Ceballos, H. G. (2019). *Intelligent Navigation of Linked Data Using Semantic Similarity.* VARE.
- Sánchez-Díaz, X., Ayala-Bastidas, G., Fonseca-Ortiz, P., et al. (2018). *A Knowledge-Based Methodology for Building a Conversational Chatbot.* MICAI (LNCS 11289).
- Fonseca, P. et al. (2025). *Improving Lean Construction by Integrating ML for Contractor Performance Prediction.* FRUCT 38.

- Fonseca, P. (2026). *Scope-Aware Contractor Performance Prediction Using ML and WBS Similarity.* FLAIRS-39 (Accepted).
- Fonseca, P. (2024). *Artificial Intelligence for Construction 4.0: Changing the Paradigms of Construction.* In M. Cebral-Loureda, E. G. Rincón-Flores, and G. Sanchez-Ante (Eds.), *What AI Can Do: Strengths and Limitations of Artificial Intelligence* (1st ed.). Taylor & Francis.

Experience

Full Time Lecturer — Computer Science 2023–Present
University of Texas Rio Grande Valley
Teaching Software Engineering, CS1 Programming, and ML-related labs.

Lecturer — Computer Science 2018–2023
Tecnológico de Monterrey
Courses in programming, digital arts, innovation, and emerging technologies.

Technology Director 2017–2021
Bohrim Consulting
Led development of construction quality systems, audit platforms, and ML-driven analytics for contractor performance.

Leader of Strategic Projects of the Provost Office 2021–2023
Tecnológico de Monterrey
Managed institutional academic initiatives, budgets, and technology implementation.

EdTech Open Innovation Coordinator 2018–2020
Tecnológico de Monterrey
Coordinated international innovation programs with MIT SOLVE and partner universities.

Technical Skills

Programming: Python, C++, JavaScript
Machine Learning: Scikit-learn, XGBoost, TensorFlow, NLP (Sentence Transformers)
Tools: Git, Docker, Django, Unity
Data & Geo: Pandas, Rasterio, GeoPandas, Photogrammetry workflows

Awards

RoboBoat Special Award (AUVSI)	2017
Graduate Scholarship — Tecnológico de Monterrey	2016
National Research Grant — CONACyT	2016
3rd Place — Hack MTY	2016
OECD Social Technology Inclusion Award	2016
OECD Disney Innovation Award	2016

Languages

Spanish (Native)
English (Fluent)